សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ | Royal University of Phnom Penh
មហាវិទ្យាល័យវិស្វកម្ម | Faculty of Engineering
ដេប៉ាតេម៉ង់ វិស្វកម្មបច្ចេកវិទ្យាព័ត៌មាន | Department of Information Technology Engineering

FULL-STACK AI ENGINEERING PROPOSAL

KhmerDoc: AI Document Management & Structured OCR

Asynchronous OpenCV preprocessing, Khmer text recognition, and human-in-the-loop verification workbench.

JUNIOR PROJECT PRACTICUM

PRESENTED BY
Po Phearun

DEGREE PROGRAM
B.Eng. in Information Technology Engineering

CORE STACK
React · Laravel · FastAPI · OpenCV

01 / PROBLEM STATEMENT

Why Khmer Document Digitization Fails

CORE BOTTLENECK

Paper Scanned Forms Trap Critical Data

Manual transcription of Khmer administrative forms is slow. Raw OCR tools dump unstructured text without field mapping.

Complex Multi-Tier Script

Base consonants stack with coeng subscripts, upper/lower vowels, and diacritics.

Unicode Encoding Ambiguity

Identical visual syllables suffer from inconsistent vowel ordering and hidden zero-width spaces.

Mobile Capture Distortion

Smartphone scans introduce perspective skew, uneven shadows, and sensor grain.

Concrete System Deliverables

Async DMS Platform

Non-blocking Redis queue processing.

FastAPI OCR Service

Isolated OpenCV enhancement microservice.

Field Extraction

Automated Khmer JSON key-value parsing.

Review Workbench

Split-screen human verification interface.

From Scanned Khmer Form to Verified JSON

INPUT: PRINTED ADMINISTRATIVE FORM

ពាក្យស្នើសុំរដ្ឋបាល

គោត្តនាមនិងនាម៖ សុខ ដារ៉ា

ភេទ៖ ប្រុស

ថ្ងៃខែឆ្នាំកំណើត៖ ១៤/០៨/២០០៣

ទីលំនៅ៖ ខណ្ឌទួលគោក រាជធានីភ្នំពេញ

OUTPUT: STRUCTURED API PAYLOAD


```
{ "doc_type": "admin_request_form", "full_name_kh": { "value": "សុខ ដារ៉ា", "confidence": 0.94,
"status": "verified" }, "dob": { "value": "១៤/០៨/២០០៣", "confidence": 0.91, "status": "verified" }
}
```

04 / SYSTEM ARCHITECTURE

Modular Monolith + Isolated AI Microservice

1. React UI
Scanner & Review UI

2. Laravel API
Sanctum Auth & Storage

3. Redis Queue
Async Job Worker

4. FastAPI OCR
OpenCV + Khmer Model

5. MySQL 8
utf8mb4 Relational DB

Architectural Decision: Asynchronous Redis jobs prevent HTTP timeouts during heavy OpenCV & OCR inference.

05 / AI & IMAGE PROCESSING PIPELINE

Inside the Python FastAPI Microservice

1. Denoising & Grayscale

OpenCV Fast Non-Local Means removes mobile camera grain.

2. Deskew & Binarization

Contour angle correction plus Adaptive Gaussian Thresholding.

3. Khmer OCR Inference

Extracts Khmer lines, bounding box coordinates, and confidence.

4. Unicode NFC & Regex

Strips zero-width spaces and matches Khmer field anchors.

Evaluating Preprocessing Impact on Khmer CER

Experiment ID	Preprocessing Configuration	Primary Evaluation Goal
Experiment A	Raw Mobile Camera Image (Baseline)	Measure baseline Character Error Rate
Experiment B	Deskewing Only	Isolate impact of rotational alignment
Experiment C	Denoising + Adaptive Thresholding	Isolate shadow and noise removal
Experiment D	**Combined OpenCV Pipeline (Proposed)**	**Maximize CER & Field Extraction Accuracy**

Primary Recognition Metric: Character Error Rate

Substitutions (S), Deletions (D), Insertions (I) over total ground-truth characters (N).

$$\mathrm{CER} = \frac{S + D + I}{N} \times 100\%$$

07 / EXECUTION ROADMAP & CURRENT STATUS

16-Week Plan & Working Prototype

WEEKS 1–4

Foundation & DB

SRS, 8-table MySQL ERD, Laravel API.

WEEKS 5–8

OCR Microservice

FastAPI, OpenCV pipeline, Redis queue.

WEEKS 9–12

Review Workbench

Split-screen UI, Regex field extractor.

WEEKS 13–16

CER Benchmark

Ablation experiments & Docker deploy.

CURRENT PROGRESS: READY FOR ADVISOR DEMO

8-table database schema and FastAPI + React OCR Workbench prototype built.

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